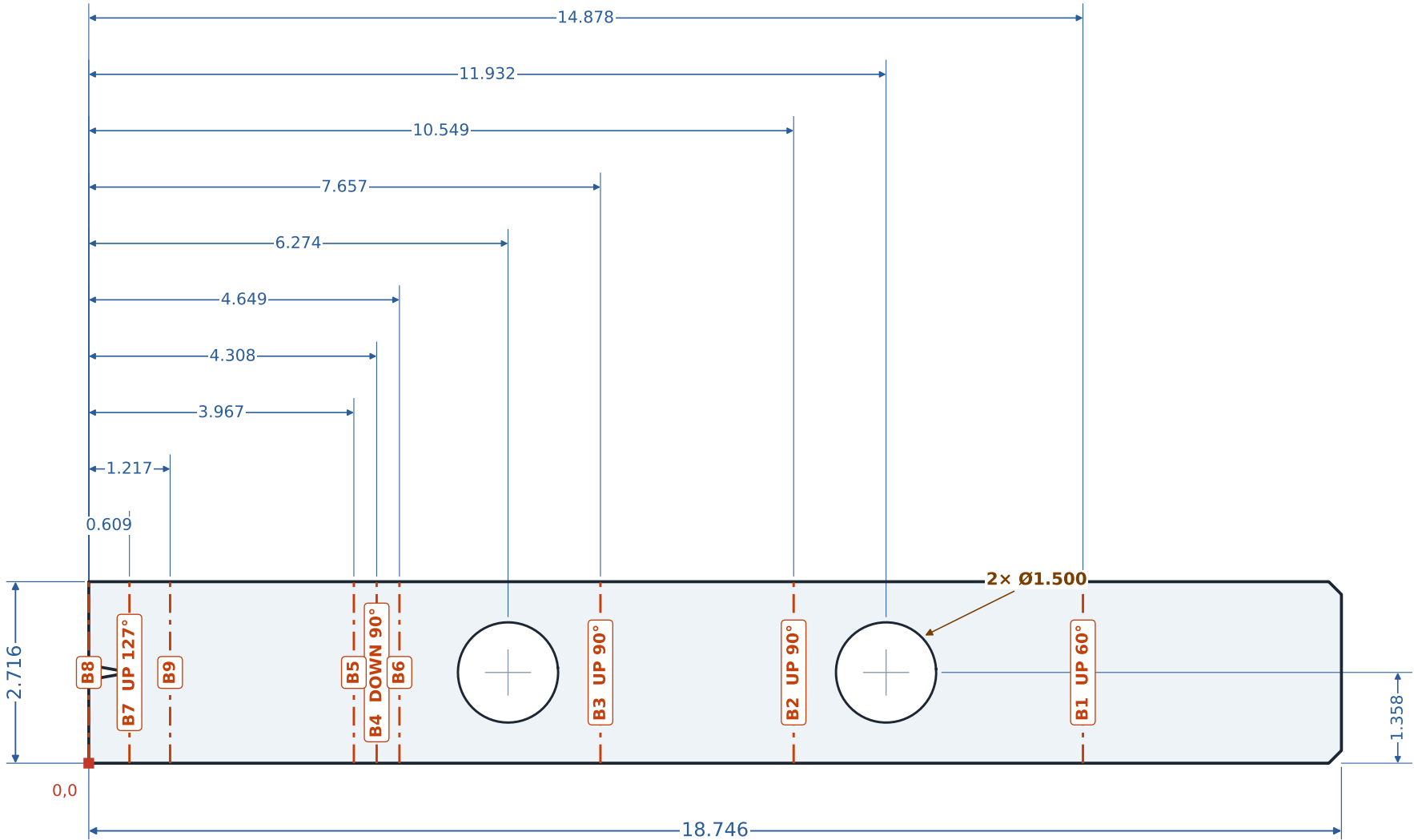


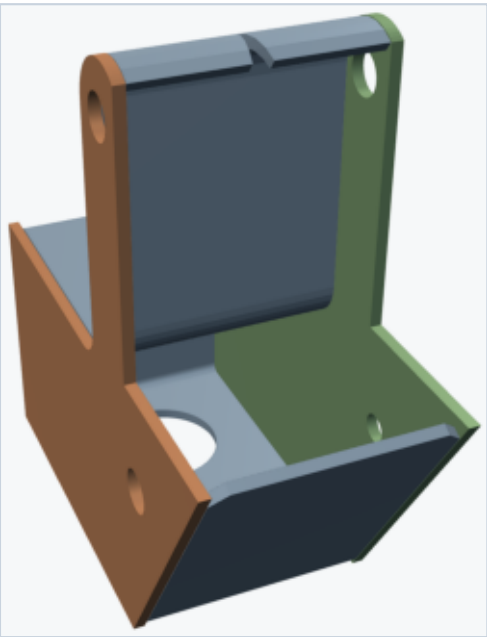
FR-4-1 · Wrap strip (ear, top, front, bottom, slant)

10 ga 0.135" (3.42 mm) · HR steel sheet A1011 CS · flat blank 18.746" × 2.716" · ≈ 1.80 lb each · QTY 1 FR + 1 FL = 2 per vehicle

FLAT PATTERN (as laser-cut, marked side up)



FORMED PART (after bending)



BEND TABLE

Bend	Angle	Direction	Inside R	Order	Bend line position
B1	60.0°	UP (toward)	R=t 0.135	5	X = 14.878
B2	90.0°	UP (toward)	R=t 0.135	4	X = 10.549
B3	90.0°	UP (toward)	R=t 0.135	3	X = 7.657
B4	90.0°	DOWN (away)	R 0.380	2	X = 4.308
B5	—	radius zone edge	—		X = 3.967
B6	—	radius zone edge	—		X = 4.649
B7	126.9°	UP (toward)	R 0.490	1	X = 0.609
B8	—	radius zone edge	—		X = 0.000
B9	—	radius zone edge	—		X = 1.217

HOLES (centre from 0,0)

Hole	Size	X	Y	Made by
H1	Ø1.500	6.274	1.358	laser
H2	Ø1.500	11.932	1.358	laser

FILES

- FR-4-1_FR_cut.dxf (right, qty 1 / vehicle)
- FR-4-1_FL_cut.dxf (left = mirror, qty 1 / vehicle)
- FR-4-1_ref.dxf (reference: + bend lines, drill-later)

NOTES

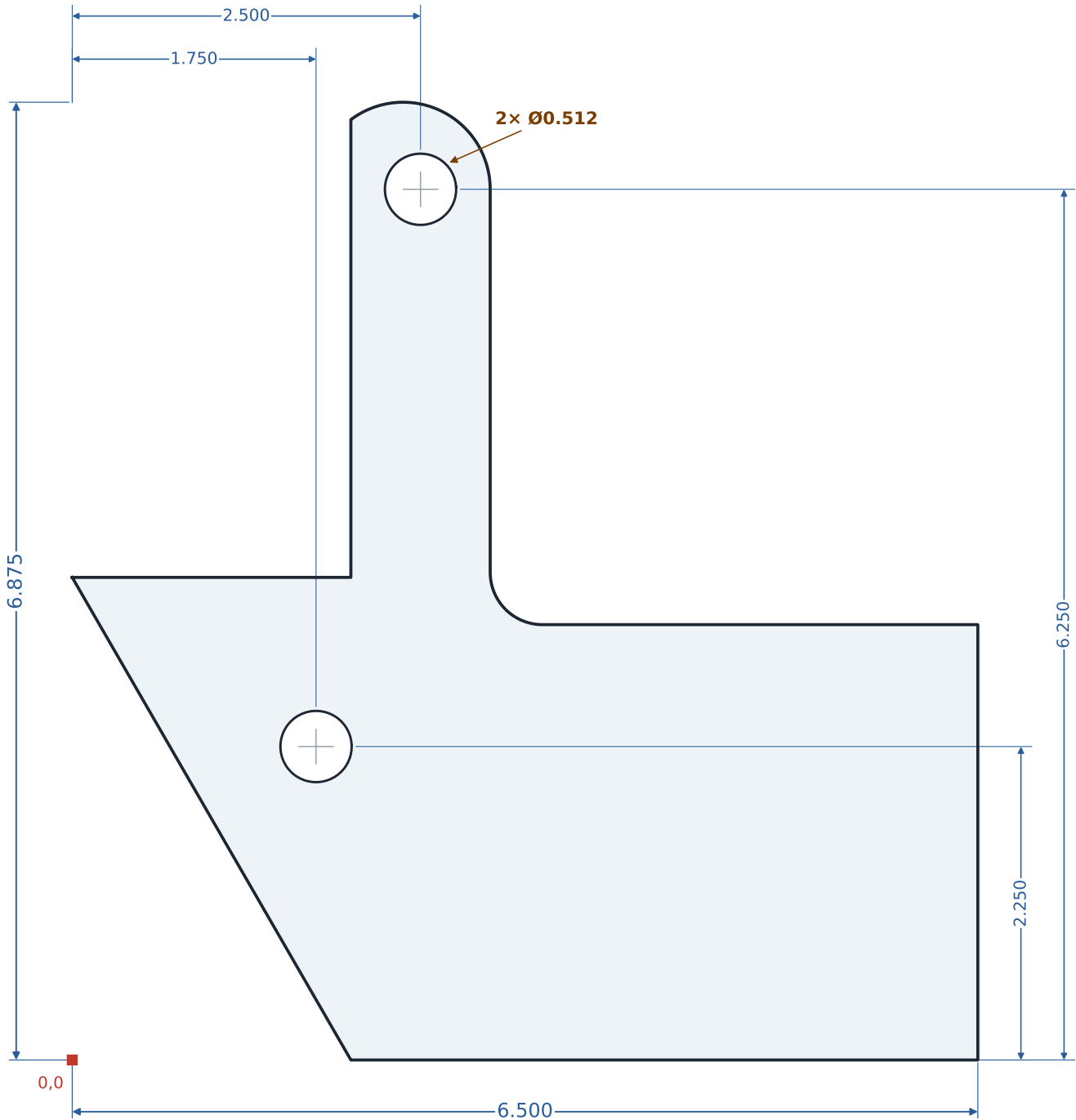
- Laser cut from the *_cut.dxf file (1:1, inch). The DXF governs the geometry; dimensions here are for checking.
 - Bend allowance (K = 0.44, inside radius = t) is already in the flat. Bend on the orange lines; UP = toward you when the part lies as drawn.
 - Holes marked DRILL LATER (red dashed) are not laser cut — drill after bending/welding.
 - Deburr all edges, remove dross.
 - Right part (FR) shown. Left part (FL) = mirror: cut *_FL_cut.dxf, bend the same UP/DOWN.
- * arc C-D bent over R 0.48; R ⅜" at E; no corner reliefs

FR-4	Bracket: wrap strip + 2 side plates + tube	HR steel sheet A1011 CS 10 ga 0.135" (3.42 mm)	Units: inch · NOT TO SCALE	Sienna WAV frame	REV A
FR-4-1 · Wrap strip (ear, top, front, bottom, slant)	QTY 1 FR + 1 FL = 2 per vehicle		Tol.: cut ±0.010" · bend ±1° · IR = t	2026-10-03	SHEET 1/4

FR-4-2 · Side plate A (2 holes Ø0.512)

10 ga 0.135" (3.42 mm) · HR steel sheet A1011 CS · flat blank 6.500" × 6.875" · ≈ 0.79 lb each · QTY 1 FR + 1 FL = 2 per vehicle

FLAT PATTERN (as laser-cut, marked side up)



NOTES

- 1. Laser cut from the *_cut.dxf file (1:1, inch). The DXF governs the geometry; dimensions here are for checking.
 - 2. Bend allowance (K = 0.44, inside radius = t) is already in the flat. Bend on the orange lines; UP = toward you when the part lies as drawn.
 - 3. Holes marked DRILL LATER (red dashed) are not laser cut — drill after bending/welding.
 - 4. Deburr all edges, remove dross.
 - 5. Right part (FR) shown. Left part (FL) = mirror: cut *_FL_cut.dxf, bend the same UP/DOWN.
- * flat, no bends

HOLES (centre from 0,0)

Hole	Size	X	Y	Made by
H1	Ø0.512	2.500	6.250	laser
H2	Ø0.512	1.750	2.250	laser

FILES

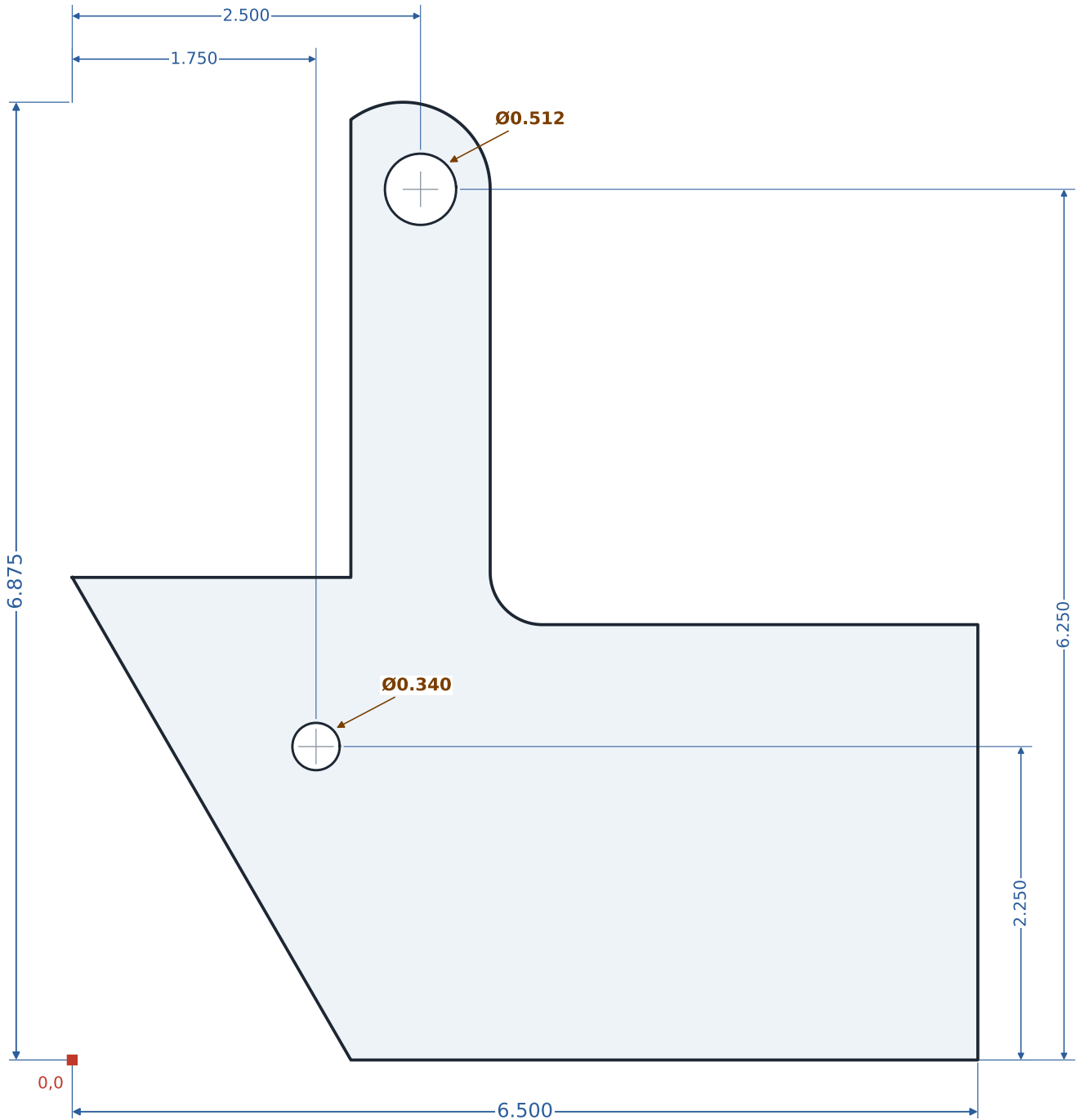
- FR-4-2_FR_cut.dxf (right, qty 1 / vehicle)
- FR-4-2_FL_cut.dxf (left = mirror, qty 1 / vehicle)

FR-4	Bracket: wrap strip + 2 side plates + tube	HR steel sheet A1011 CS 10 ga 0.135" (3.42 mm)	Units: inch · NOT TO SCALE	Sienna WAV frame	REV A
FR-4-2 · Side plate A (2 holes Ø0.512)		QTY 1 FR + 1 FL = 2 per vehicle	Tol.: cut ±0.010" · bend ±1° · IR = t	2026-10-03	SHEET 2/4

FR-4-3 · Side plate B (holes Ø0.512 + Ø0.34)

10 ga 0.135" (3.42 mm) · HR steel sheet A1011 CS · flat blank 6.500" × 6.875" · ≈ 0.80 lb each · QTY 1 FR + 1 FL = 2 per vehicle

FLAT PATTERN (as laser-cut, marked side up)



NOTES

- 1. Laser cut from the *_cut.dxf file (1:1, inch). The DXF governs the geometry; dimensions here are for checking.
 - 2. Bend allowance (K = 0.44, inside radius = t) is already in the flat. Bend on the orange lines; UP = toward you when the part lies as drawn.
 - 3. Holes marked DRILL LATER (red dashed) are not laser cut — drill after bending/welding.
 - 4. Deburr all edges, remove dross.
 - 5. Right part (FR) shown. Left part (FL) = mirror: cut *_FL_cut.dxf, bend the same UP/DOWN.
- * flat, no bends

HOLES (centre from 0,0)

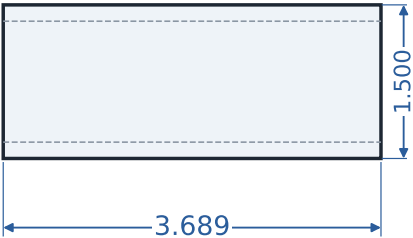
Hole	Size	X	Y	Made by
H1	Ø0.512	2.500	6.250	laser
H2	Ø0.340	1.750	2.250	laser

FILES

- FR-4-3_FR_cut.dxf (right, qty 1 / vehicle)
- FR-4-3_FL_cut.dxf (left = mirror, qty 1 / vehicle)

FR-4	Bracket: wrap strip + 2 side plates + tube	HR steel sheet A1011 CS 10 ga 0.135" (3.42 mm)	Units: inch · NOT TO SCALE	Sienna WAV frame	REV A
FR-4-3 · Side plate B (holes Ø0.512 + Ø0.34)	QTY 1 FR + 1 FL = 2 per vehicle	Tol.: cut ±0.010" · bend ±1° · IR = t	2026-10-03	SHEET 3/4	

FR-4 · tube cut list



Tube OD 1.5 / ID 1.181 — cut 3.689" (through bottom and top)

Stock: DOM round steel tube OD 1.500" × ID 1.181" (wall ≈ 0.160")

Cut length 3.689" ±1/32", square ends, deburr

QTY 1 per side × 2 (FR + FL) = 2 per vehicle

FR-4	Bracket: wrap strip + 2 side plates + tube	see stock above	Units: inch · NOT TO SCALE	Sienna WAV frame	REV A
Tube cut list		see quantities above	Tol.: cut ±0.010" · bend ±1° · IR = t	2026-10-03	SHEET 4/4